



OFMFS: Optimizing feature diversity via multi-perspective self-supervised tasks for few-shot image classification

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ABSTRACT

Few-shot learning (FSL) focuses on recognizing novel categories from only a small number of labeled examples. Recently, feature optimization has enhanced model classification by optimizing feature extraction and representation. However, existing methods pay insufficient attention to frequency-domain information, and their feature representations may remain sensitive to coarse visual patterns when only a few support samples are available. In this paper, we propose an optimizing feature diversity via multi-perspective self-supervised tasks (OFMFS), which learns multi-granularity feature representations through two complementary stages. OFMFS comprises feature multi-perspective training (FMT) and feature diversity learning (FDL). The FMT strategy enforces the model to predict image rotation angles and corresponding wavelet subbands, enabling joint exploration of geometric structures and frequency-domain details. The FDL module enhances sensitivity to fine-grained features by randomly swapping patches between samples to break hidden feature dependencies. The evaluation on three FSL benchmarks confirms the effectiveness of OFMFS. Under the 5-way 5-shot setting, OFMFS achieves accuracy gains of 4.99% on CUB-200-2011 and 3.27% on MiniImageNet over the baseline.

1. Introduction

Few-shot learning (FSL) focuses on recognizing novel categories from only a few labeled examples by exploiting transferable knowledge from base classes [1]. Although traditional deep learning algorithms perform well in scenarios with adequate training data, they face significant challenges in data-scarce situations [2]. This challenge is especially evident in areas like biology and medicine [3]. Recent AI-driven triage and clinical decision-support studies confirm this need. They explore foundation models, explainable medical image analysis networks, and augmented-reality decision-support systems for disease detection, tumor segmentation, and multimodal clinical data interpretation [4–6]. Unlike conventional deep learning methods, FSL learns transferable representations that can be adapted to novel tasks with limited training data. Currently, FSL has made remarkable progress in various areas, including medicine and bioinformatics [7], industrial and engineering applications [8], and natural language processing [9].

Recent FSL methods are commonly studied from three perspectives: meta-learning, feature optimization, and self-supervised learning approaches. Meta-learning simulates numerous tasks during training to acquire shared knowledge among tasks. These methods learn task-level priors so that the model can generalize to unseen tasks with limited supervision. Although meta-learning methods demonstrate strong adaptability in FSL, they have limitations when feature representations are insufficient. Unlike meta-learning, feature optimization methods enhance class separability and consistency by designing more discriminative feature extractors or using

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generative models to expand data features [3]. These approaches are categorized into discriminative-based and generative-based feature optimization [10]. Discriminative feature optimization aims to learn more separable embeddings by reducing variations within the same class and enlarging differences between different classes [11]. By contrast, generative-based feature optimization expands data distribution and enriches the feature space by generating additional samples or features [12]. Although feature optimization methods improve FSL performance by enhancing feature expression and expanding data distribution, they have limitations in utilizing unlabeled data and enhancing model adaptability to unknown tasks.

Unlike feature optimization approaches, self-supervised learning tailors pre-training tasks to extract latent feature representations using unlabeled data. This paradigm enhances the robustness of feature extractors for downstream tasks by leveraging intrinsic data structures [13]. These methods can be classified into contrastive self-supervised learning and generative self-supervised learning [14]. Contrastive self-supervised learning leverages contrastive loss to achieve feature consistency among similar samples [15]. Chen et al. [15] proposed different contrastive learning frameworks, establishing important milestones for this field. Generative self-supervised learning focuses on feature generation and reconstruction by exploring data distribution through generative tasks [16]. These methods enable effective feature representation learning for downstream tasks with limited data through pre-training.

The above self-supervised methods improve representation learning for FSL tasks. However, for image-based few-shot classification, two observations still motivate further exploration: (1) Some generative self-supervised tasks mainly rely on spatial transformations, while frequency-domain details may be insufficiently exploited. As shown in Observation 1 of Fig. 1, when the model relies on self-supervised tasks involving spatial transformations such as rotation angles or image coloring, it may learn shortcuts from easily accessible visual cues. An example is the possible shortcut that “a red object means a cardinal bird”. In data-scarce scenarios, insufficient use of frequency-domain details may limit the diversity and robustness of the learned image representations. (2) In a few-shot setting, the feature information extracted by the model from limited samples may be insufficient, leading to low sensitivity to fine-grained features. As shown in Observation 2 of Fig. 1, FSL models may rely on coarse visual patterns for rough matching. For instance, they may use coarse-grained features such as “gray and large body” for comparison, underutilizing fine-grained features such as bird beaks and feather edges. When only a few samples are available in the support set, the model tends to learn the overall appearance differences between categories. This operation may make it difficult to capture the fine-grained features that truly determine the category.

To overcome these limitations, we introduce multi-granularity representation learning to combine global semantic structure and local discriminative cues. Accordingly, we propose an FSL method that optimizes feature diversity representation through multi-angle and multi-scale self-supervised tasks (OFMFS). OFMFS includes a two-stage strategy of feature multi-perspective training (FMT) and feature diversity learning (FDL). For the challenges of spatial information dependency, we design an FMT strategy that forces the model to predict the rotation angles and corresponding wavelet subbands of images. This stage task encourages the model to explore potential patterns and structures in the data from multiple angles and scales. For the challenges of low sensitivity to fine-grained features, we propose an FDL module that randomly exchanges local patch blocks in sample feature maps during downstream tasks, breaking the inherent relationships between features in hidden layers. This phase task transfers useful information between different samples to improve feature utilization. OFMFS provides more diverse feature representations for FSL tasks by integrating multi-granularity feature learning across global and local perspectives.

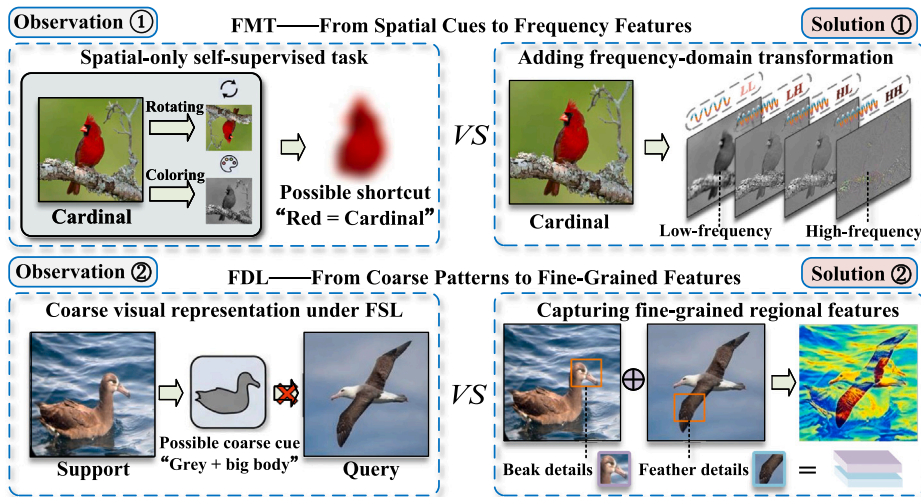


Fig. 1. Illustration of the motivation behind OFMFS for few-shot image classification. The left side shows two potential limitations that may arise in limited-data image recognition: spatial shortcut learning caused by spatial-only self-supervised tasks, and coarse visual representation caused by limited support samples. The right side illustrates the corresponding design motivations of OFMFS. FMT introduces frequency-domain supervision to complement spatial transformations, and FDL encourages patch-level feature diversity to better leverage fine-grained discriminative image regions.

We verify OFMFS on MiniImageNet, CIFAR-FS, and CUB-200-2011, covering both general and fine-grained few-shot image classification scenarios. In the 5-way 5-shot setting, OFMFS improves the baseline by 3.27%, 1.42%, and 4.99% on the three datasets, respectively. The contributions of this work are as follows:

- From the global view, we introduce a multi-perspective self-supervised task to exploit both geometric structure and frequency subband information. This task helps the model learn multi-granularity representations for few-shot image classification.
- From the local view, we introduce FDL to reorganize patch-level features in the hidden representation space. This operation encourages the model to exploit diverse local cues and improves its sensitivity to fine-grained discriminative regions.
- We reveal the synergy between FMT and FDL for few-shot image classification. FMT learns rotation-aware and frequency-aware image representations, while FDL improves local feature diversity through feature-map reorganization.
- Experiments on three representative FSL benchmarks verify the effectiveness of OFMFS and show consistent improvements over existing methods.

The remaining sections are structured as follows. Section 2 provides a concise overview of prior work related to FSL. Then, Section 3 details the OFMFS model. Section 4 reports the experimental settings, comparisons, and analyses. Section 5 presents a brief conclusion and delineates the directions for future work.

2. Related work

This section discusses prior FSL studies from two aspects: self-supervised learning and feature optimization.

2.1. FSL based on self-supervised learning

Self-supervised learning enhances the feature extraction capability for downstream tasks by designing different pretext tasks. This pre-training process hardly requires additional labeled samples. Through these tasks, the model learns the essential patterns and structure of the input data. The pre-trained feature extractor is then applied to downstream FSL tasks, improving their performance [13]. Current FSL-based self-supervised tasks are mainly categorized into contrastive and generative methods [14].

Contrastive self-supervised learning captures unique data structures by boosting positive pair similarity and diminishing negative pair similarity [15]. Ouyang et al. [17] constructed positive and negative pairs to obtain potential similarities and differences in user behaviors, improving the performance of recommendation systems. Recognizing the special temporal characteristics of time series, Liu et al. [18] adopted contrastive learning to build a universal representation for time series data. For FSL downstream tasks, Liu et al. [19] enhanced the effectiveness of embedding models by designing a specific contrastive learning framework. Extending to financial security, Kasula et al. [20] proposed the Sea Lion Self-Supervised Network, which automatically learns discriminative feature representations between normal and fraudulent transactions without requiring a large number of labeled fraud samples.

Generative self-supervised learning explores intrinsic data structures through reconstruction-based pretext tasks [15]. Gidaris et al. [16] used image rotation prediction to guide visual representation learning, allowing it to obtain more discriminative feature representations. On this basis, Mangla et al. [21] generated new intermediate representations by mixing the features in downstream tasks. Inspired by knowledge distillation, Rajasegaran et al. [13] combined self-supervised pre-training with knowledge distillation, transferring the learned representations to the few-shot model. From the frequency-domain perspective, Xie et al. [22] proposed a frequency-domain masked self-supervised task, allowing the model to reconstruct masked frequency information to obtain more powerful visual features.

Although existing self-supervised FSL methods achieve promising performance, many of them mainly rely on spatial transformations, and frequency-domain information is not sufficiently exploited in limited-data image classification. Inspired by generative self-supervised learning, we design a self-supervised strategy that predicts the rotation angles and wavelet subbands of images. This module aims to explore the inherent patterns and structural information of data from a multi-perspective, alleviating the problem of sample scarcity in FSL. Compared with the above methods, our approach further introduces frequency band prediction as an auxiliary task target in addition to predicting rotation angles.

2.2. FSL based on feature optimization

Feature optimization methods seek to improve the representation space in FSL, enabling models to obtain more discriminative features from limited samples. These methods can be classified into discriminative and generative feature optimization approaches [23].

Discriminative feature optimization-based FSL methods enhance inter-class separability and intra-class consistency by optimizing feature representations [24]. Dong et al. [11] enhanced few-shot classification capability by learning more discriminative embeddings and enlarging the separation between categories. Building on this, Perera et al. [24] guided feature adaptation using a few samples, thereby reducing the number of parameters required for model adjustment. From a multi-scale geometric structure perspective, Li et al. [25] modeled data geometric features progressively from fine-grained to coarse-grained levels, enabling the model to understand complex structural information from limited samples.

Generative feature optimization-based FSL methods expand the training dataset or enhance feature representation using generative models [26]. Zhang et al. [10] employed variational auto-encoders (VAE) to model the potential feature distribution of FSL, strengthening the generalization of the model under data scarcity. Unlike VAE-based methods, Ren et al. [27] utilized GANs to create

additional samples, addressing significant data imbalance. From the perspective of feature space, Yang et al. [28] enriched the feature space through a simplified interpolation strategy, improving representation diversity under scarce samples.

Although feature optimization methods improve representation ability in FSL, they may still be limited in capturing fine-grained features when only a few support samples are available. Unlike generative feature optimization methods, we design a feature optimization strategy for downstream tasks. This method performs interactive interpolation on random feature regions from different samples, breaking the inherent constraints between features in hidden layers.

3. The OFMFS model

This section presents the OFMFS architecture and explains its two-stage design. Next, the learning procedure and model performance are analyzed to clarify how OFMFS performs few-shot image classification.

3.1. Framework overview

OFMFS follows a progressive two-stage design to learn complementary global and local feature representations. Optimizing feature diversity via multi-perspective self-supervised tasks (OFMFS) consists of two primary stages: feature multi-perspective training (FMT) and feature diversity learning (FDL). The fundamental architecture of OFMFS is depicted in Fig. 2. Although rotation prediction, frequency-domain learning, and feature reorganization have been studied separately, their staged integration remains insufficiently explored in few-shot image classification. Therefore, the innovation of OFMFS lies in the progressive FMT-FDL framework.

(1) **FMT**: We design a self-supervised strategy that establishes dual-supervisory signals to predict image rotation angles and wavelet subband information simultaneously.

(2) **FDL**: We propose a feature optimization module that enhances feature representation by implementing patch-level interpolation in the hidden representation space.

3.2. Feature multi-perspective training

In this phase, we formulate a multi-perspective self-supervised strategy, aiming to optimize feature diversity in data-limited scenarios through geometric invariance and frequency-aware discriminability theories. Geometric invariance guides the model in preserving structural coherence across geometric transformations. This theory enables the network to decouple rotation information from related semantics, maintaining stable feature representations under spatial transformations. As a complement, frequency-aware discriminability precisely identifies wavelet decomposition subbands. This acts as a spectral regularizer to separate the low-frequency

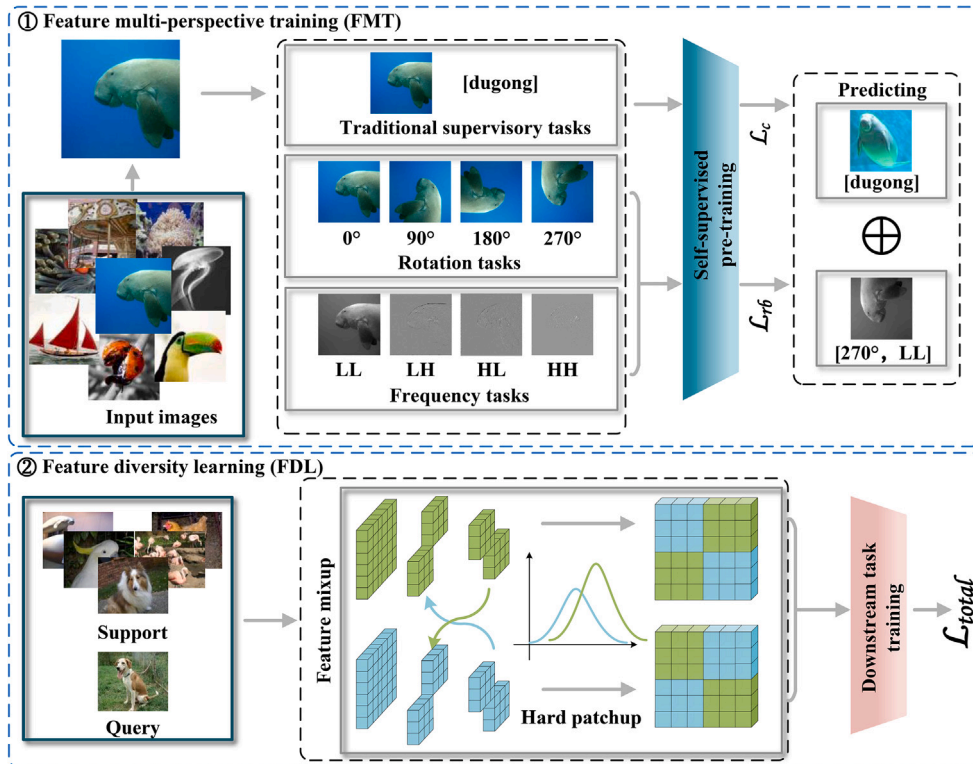


Fig. 2. The framework of OFMFS.

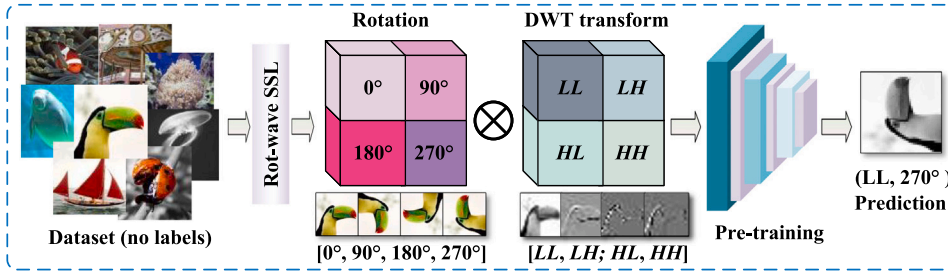


Fig. 3. Specific steps demonstrating the self-supervised strategy in FMT stage.

overall structure and high-frequency texture details of images. This approach establishes a connection between spatial transformation and frequency band discrimination, filling the semantic gap in feature embedding spaces of FSL.

The FMT stage adopts a dataset configured as follows. The dataset consists of samples paired with their labels, used for the FMT phase. Specifically, the sample set $\mathbf{D} = \{(x_i, y_i)\}_{i=1}^m$, where x_i denotes the i -th sample, y_i represents the label of sample x_i , and m indicates the total number of samples. We utilize angle rotation and two-dimensional discrete wavelet transform (DWT) for the secondary processing of images. DWT analyzes the image at multi-scales from the frequency-domain perspective and captures the different frequency information in the image. In our work, r signifies the specific random rotation applied to an image and b refers to the identity of DWT applied to the image. Additionally, rb represents the identity of the random rotation and DWT applied to the image simultaneously. The set $\mathbf{C}^r$ encompasses all rotation angles, formulated as $\mathbf{C}^r = \{r_j\}_{j=1}^{p_r}$, where r_j denotes the j -th rotation angle. This set includes 0° , 90° , 180° , and 270° . The set $\mathbf{C}^b$ consists of wavelet subbands represented as $\mathbf{C}^b = \{b_n\}_{n=1}^{p_b}$, where b_n denotes the n -th wavelet subband. This includes subbands such as LL , LH , HL , and HH . The set $\mathbf{C}^{rb}$ is formulated as $\mathbf{C}^{rb} = \{z_k^{rb}\}_{k=1}^d$, where each z_k^{rb} represents the k -th joint self-supervised pseudo-label consisting of a rotation angle and a wavelet subband identity. Specifically, $\mathbf{C}^{rb}$ is constructed by randomly pairing elements from the rotation angle set $\mathbf{C}^r$ and the wavelet subband set $\mathbf{C}^b$, resulting in $\{(0^\circ; LL), \dots, (270^\circ; HH)\}$. The set $\mathbf{D}^r$ represents the result of randomly rotating the dataset $\mathbf{D}$ by four different angles. Each rotated sample $x_i^r \in \mathbf{D}^r$ corresponds to a rotation angle r . The set $\mathbf{D}^{rb}$ consists of samples undergoing rotation and DWT. Each sample $x_i^{rb} \in \mathbf{D}^{rb}$, which is rotated and decomposed, corresponds to the combined rotation and DWT label set rb in the FMT stage. The methodology enables systematically analyzing the invariance of the dataset under multi-angle geometric transformations and multi-scale wavelet decompositions.

Rotation and DWT are applied to each image to construct joint supervision signals from geometric and frequency-domain information. Fig. 3 illustrates this process.

We standardize the images and divide them into the RGB color space. Then, we apply rotation transformations of 0° , 90° , 180° , and 270° to the input images to generate augmented samples.

The image after rotation is denoted as x_i^r , and the specific operation is

$$x_i^r = f_{rot}(x_i), \quad (1)$$

where $f_{rot}(\cdot)$ denotes the random rotation operation performed on an image, with the corresponding label being r_j .

We employ an adaptive DWT strategy for each image that is rotated at multiple angles. Through this strategy, each rotated image is decomposed into LL , LH , HL , and HH subbands, which describe low-frequency structure, horizontal details, vertical details, and diagonal details, respectively. The DWT process is determined based on the edge features E and variance σ of the image, enabling dynamic wavelet basis selection. The calculation of edge features E is

$$E = \frac{1}{2} (\text{mean}(|\nabla_x \otimes x_i^r|) + \text{mean}(|\nabla_y \otimes x_i^r|)), \quad (2)$$

where $\text{mean}(\cdot)$ denotes arithmetic mean operation, ∇_x and ∇_y are horizontal and vertical gradient calculations of the Sobel operator, $|\cdot|$ is the absolute value operation, and $\otimes$ represents convolution operations.

The calculation of variance σ is

$$\sigma = \text{Var}(x_i^r), \quad (3)$$

where $\text{Var}(\cdot)$ denotes data variance operation.

We dynamically select different wavelet bases for DWT in the face of varying edges and textures across different images. The wavelet basis parameter is denoted by ψ , and the specific selection is as

$$\psi = \begin{cases} db4, & \text{if } E > 0.2 \text{ and } \sigma > 0.15; \\ sym4, & \text{if } E > 0.2 \text{ and } \sigma \leq 0.15; \\ coif3, & \text{if } E \leq 0.2 \text{ and } \sigma > 0.1; \\ haar, & \text{otherwise,} \end{cases} \quad (4)$$

where $db4$, $sym4$, $coif3$, and $haar$ represent the options available depending on the specific circumstances. The $db4$ wavelet basis suits scenarios with high edge intensity and complex textures. The $sym4$ wavelet basis is appropriate for high edge intensity but smooth

texture scenarios. The *coif3* wavelet basis is appropriate for scenarios with low edge intensity but rich details. The *haar* wavelet basis is selected in the remaining case, where the edge response and texture complexity are both relatively low. The *haar* wavelet basis has the shortest support and a simple step-like basis, capturing coarse structural changes with low computational cost. Therefore, *haar* wavelet basis is suitable for structurally simple images and avoids introducing redundant high-frequency responses from more complex wavelet bases.

After selecting the appropriate wavelet basis, we perform DWT on the rotated images to extract the *LL*, *LH*, *HL*, and *HH* subbands. The specific operation of the adaptive wavelet transform function $DWT_{\psi}(\cdot)$ is

$$DWT_{\psi}(x_i^r) = [LL, LH; HL, HH]. \quad (5)$$

Each subband is upsampled and concatenated to restore the original image resolution $H \times W$. After rotation and DWT, the sample x_i^{rb} is

$$x_i^{rb} = \text{Stk}(\text{Up}(LL), \text{Up}(LH), \text{Up}(HL), \text{Up}(HH)), \quad (6)$$

where $\text{Stk}(\cdot)$ is a dimension stacking operation, and $\text{Up}(\cdot)$ is an upsampling operation.

After the joint labels are obtained, we can feed the processed samples into the network to optimize the self-supervised task in the FMT stage. The self-supervised loss of the FMT phase $\mathcal{L}_{rb}$ is

$$\mathcal{L}_{rb} = \frac{1}{m \cdot d} \sum_{i=1}^m \sum_{k=1}^d \mathcal{L}(K_{rb}(f_{\theta}(x_i^{rb})), z_k^{rb}), \quad (7)$$

where $\mathcal{L}(\cdot, \cdot)$ denotes the cross-entropy loss, $K_{rb}(\cdot)$ is a classifier for the multi-perspective self-supervised task, and $f_{\theta}(\cdot)$ is the feature extraction network. Here, the logits are calculated using scaled cosine similarity between the feature vector and each class weight.

The cross-entropy loss $\mathcal{L}_c$ is designed to predict the true class labels of input images, and its formulation is

$$\mathcal{L}_c = \mathcal{L}(K_c(f_{\theta}(x_i)), y_i), \quad (8)$$

where $K_c(\cdot)$ is the classifier that predicts the true label and y_i is the true class label. $\mathcal{L}_c$ serves as a semantic anchor to preserve class-level discriminability. This operation prevents the feature extractor from being dominated solely by the multi-perspective pseudo-label prediction task.

The total loss formulation for the FMT stage is

$$\mathcal{L}_{FMT} = \alpha \mathcal{L}_c + (1 - \alpha) \mathcal{L}_{rb}, \quad (9)$$

where α adaptively adjusts the influence of the classification loss $\mathcal{L}_c$ and the self-supervised loss $\mathcal{L}_{rb}$, reducing possible conflicts between semantic classification and multi-perspective prediction.

The unbounded adaptive weight parameter $\tilde{\alpha}$ is adjusted based on the following strategy:

$$\tilde{\alpha} = Ep \cdot \gamma \cdot \left(1 + \frac{\mathcal{L}_{rb}}{\mathcal{L}_c}\right) \cdot \left(1 - \frac{|\mathcal{A}_{rb} - \mathcal{A}_c|}{100}\right), \quad (10)$$

where $\mathcal{A}_{rb}$ and $\mathcal{A}_c$ denote the accuracies of the multi-perspective prediction and class-label prediction tasks, respectively. Ep is the current epoch and γ controls the relationship of batches to the adaptive weight parameter α . In our work, γ is set to 0.003.

The final adaptive weight parameter α representation is

$$\alpha = \text{clip}(\tilde{\alpha}, 0, 1), \quad (11)$$

where $\text{clip}(\cdot, 0, 1)$ is the explicit projection operation that projects $\tilde{\alpha}$ into the interval $[0, 1]$. This adaptive weighting strategy is an empirical balancing strategy.

In the initial stages of training, the parameter α focuses on the standard classification task to stabilize feature identification. As the number of batches increases, the parameter α is reduced. At the same time, we enhance the self-supervised task to constrain feature diversity. This process ultimately leads to the collaborative optimization of the dual tasks.

Through this stage, the model acquires coarse-grained global priors that serve as anchors for subsequent fine-grained feature learning.

3.3. Feature diversity learning

In the N -way K -shot setting, the training data are organized into base classes $\mathbf{D}_b = \{(x_i, y_i)\}_{i=1}^{m_b}$ and novel classes $\mathbf{D}_n = \{(x_i, y_i)\}_{i=1}^{m_n}$, where m_b and m_n denote the numbers of base and novel samples, respectively. The model first learns transferable representations from the labeled base classes $\mathbf{D}_b$ and is then evaluated on novel classes $\mathbf{D}_n$ with only a few labeled examples.

During the FDL stage, we employ the PatchUp technique for dynamic feature restructuring to optimize feature utilization for FSL. Specifically, we divide the original image of each sample into multiple $N \times N$ patches. For a mini-batch containing m training samples, we collect the selected inter-layer feature maps as $\mathbf{F} = [\mathbf{F}_1; \dots; \mathbf{F}_i; \dots; \mathbf{F}_m]$, where $\mathbf{F}_i$ denotes the feature map of the i -th sample. This operation facilitates subsequent region swapping operations, where $N = \lfloor \frac{H}{block_{size}} \rfloor \times \lfloor \frac{W}{block_{size}} \rfloor$. The parameter $block_{size}$ controls the patch size and balances the capture of local details and global semantics.

For each feature map $\mathbf{F}_i$ in the mini-batch, a binary mask $\mathbf{M}_i$ is generated via the Bernoulli distribution, expressed as

$$\mathbf{M}_i \sim \text{Bernoulli}(p), \quad (12)$$

where $\text{Bernoulli}(\cdot)$ is the Bernoulli distribution.

The activation probability p is controlled by the parameter ϵ , which is formulated as

$$p = \epsilon \cdot \frac{H^2}{\text{block}_{size}^2 \cdot (H - \text{block}_{size} + 1)^2}, \quad (13)$$

where ϵ is a preset mixing intensity parameter. The denominator term is used to compensate for the scale bias of the feature map across different layers.

Next, we perform a maximum pooling operation on the obtained mask to generate the selected region. The optimized mask $\mathbf{H}$ is

$$\mathbf{H} = \text{MaxP}(\mathbf{M}), \quad (14)$$

where $\text{MaxP}(\cdot)$ is the maximum pooling operation. This operation ensures that the exchange region is a contiguous block and avoids destroying the consistency of the local semantics.

After determining the exchanged regions, we randomly select two inter-layer feature maps $\mathbf{F}_i$ and $\mathbf{F}_j$ from the mini-batch, whose true labels of $\mathbf{F}_i$ and $\mathbf{F}_j$ are y_i and y_j , respectively. We replace the empty regions of $\mathbf{F}_j$ with the corresponding blocks of $\mathbf{F}_i$ to generate a new inter-layer feature map $\mathbf{F}_k$. The new inter-layer feature map $\mathbf{F}_k$ is

$$\mathbf{F}_k = \mathbf{H} \odot \mathbf{F}_i + (\mathbf{1} - \mathbf{H}) \odot \mathbf{F}_j, \quad (15)$$

where $\odot$ denotes element-by-element multiplication.

Finally, the loss at the FDL stage is

$$\mathcal{L}_{FDL} = \text{Mix}_{\mu}(\mathcal{L}(f_{\theta}(\mathbf{F}_k), y_i), \mathcal{L}(f_{\theta}(\mathbf{F}_k), y_j)) + \mathcal{L}(f_{\theta}(\mathbf{F}_k), \text{Mix}_{\mu}(y_i, y_j)), \quad (16)$$

where y_i and y_j are the true labels of the inter-layer feature maps $\mathbf{F}_i$ and $\mathbf{F}_j$, respectively. The $\text{Mix}_{\mu}(\cdot, \cdot)$ is the mixing operation.

The total loss function can be expressed as

$$\mathcal{L} = \xi \mathcal{L}_{FMT} + (1 - \xi) \mathcal{L}_{FDL}, \quad (17)$$

where ξ adjusts the relative weights of $\mathcal{L}_{FMT}$ and $\mathcal{L}_{FDL}$ in the overall objective.

This stage complements the coarse-grained features from FMT by focusing on fine-grained local discriminative patterns, thereby achieving a comprehensive multi-granularity representation.

3.4. Model analysis and learning

OFMFS integrates self-supervised tasks to enhance FSL performance. Algorithm 1 outlines the model training pseudocode. In the FMT phase, as shown in lines 3 and 4, the model performs multi-angle image rotation and DWT to extract multi-scale wavelet subband information. By predicting rotation angles and frequency bands, OFMFS learns diverse features, enhancing feature extraction

Algorithm 1 OFMFS.

Input: The base training set $\mathbf{D}_b$ in each episode, with N classes and K support examples for each class. The two-stage training epoch numbers K_1 and K_2 .

Output: The best parameter set θ^{FDL} of OFMFS model.

```

1: Initialize feature extractor  $f_{\theta}(\cdot)$ ;
  //Feature multi-perspective training stage
2: for  $epoch \leq K_1$  do
3:   Rotate the input image by four angles by Eq. (1);
4:   Perform DWT on the rotated image by Eq. (5);
5:   Calculate the loss in the FMT stage by Eq. (9);
6:   Update feature extractors  $f_{\theta}(\cdot)$ ;
7: end for
8: Obtain the FMT-pretrained parameters  $\theta^{\text{FMT}}$ ;
  //Feature diversity learning stage
9: Initialize the FDL stage with  $\theta^{\text{FMT}}$ ;
10: for  $K_1 < epochs \leq K_2$  do
11:   Acquire the exchange region by Eq. (14);
12:   Reorganize fine-grained features by Eq. (15);
13:   Calculate the loss in FDL stage by Eq. (16);
14:   Update  $\theta$  from  $\theta^{\text{FMT}}$  to  $\theta^{\text{FDL}}$  by minimizing  $\mathcal{L}$ ;
15: end for
16: return  $\theta^{\text{FDL}}$ .
```

for the next stage. In the FDL phase, as depicted in lines 11 and 12, the model acquires the exchange region and reorganizes fine-grained features to break the inherent feature relationships and enhance feature utilization. The loss function guides model parameter optimization, with the final parameter set θ^{FDL} returned in line 16. It should be noted that the feature extractor is not frozen during the FDL stage. After the FMT stage, the learned parameter set θ^{FMT} is used to initialize the FDL stage.

OFMFS differs from conventional FSL methods in two main aspects. First, OFMFS designs self-supervised modules at the FMT stage to predict image angles and frequency subbands. This stage separates geometric features from frequency-domain information, allowing the following FDL stage to start with more structured feature representations. Second, in the FDL phase, feature diversity is further enhanced by reorganizing the fine-grained features and breaking the inherent feature relationships. This two-stage strategy enhances the diversity of feature representations and improves the adaptability of the model to new tasks.

In addition, OFMFS reduces the risk of overfitting through two mechanisms. The FMT stage introduces auxiliary rotation angles and wavelet subbands prediction tasks to increase supervision diversity without additional manual labels. The FDL stage performs patch-level feature reorganization in the hidden feature space, which acts as feature-level regularization and prevents the model from memorizing fixed local patterns.

4. Experimental results and analysis

This section reports the experimental evaluation of OFMFS, including dataset settings, implementation details, comparisons with representative methods, ablation studies, wavelet subband analysis, parameter sensitivity, visualization results, and computational cost analysis.

4.1. Datasets and experimental setup

OFMFS is evaluated on three benchmark datasets, including MiniImageNet, CUB-200-2011, and CIFAR-FS. MiniImageNet contains 100 categories, and each category has 600 images with a resolution of 84×84 . Following the standard split, we use 64, 16, and 20 classes for training, validation, and testing, respectively. CUB-200-2011 contains 11,788 images from 200 bird species, which are divided into 130 training classes, 20 validation classes, and 50 testing classes. CIFAR-FS also consists of 100 categories. We divide it into 64 training classes, 16 validation classes, and 20 testing classes, and resize all images to 32×32 .

For implementation, OFMFS adopts a two-stage training scheme. The adaptive parameter is denoted as α , and the other hyperparameters are set as $\epsilon = 0.9$, $block_{size} = 7$, and $\xi = 0.5$. The FMT and FDL stages are trained for 200 epochs each, corresponding to $K_1 = 200$ and $K_2 = 400$. All experiments are implemented with PyTorch on an NVIDIA GeForce RTX 4090D GPU and a 3.60 GHz Intel CPU.

4.2. Comparison with representative FSL methods

We evaluate OFMFS on three benchmarks using multiple methods. S2M2_R [21] serves as the baseline, and the best results are highlighted in bold.

As illustrated in Table 1, we evaluate the performance of OFMFS on the CUB-200-2011.

(1) Table 1 reveals that OFMFS improves CUB-200-2011, elevating 5-way 1-shot accuracy by 4.37% and 5-way 5-shot accuracy by 4.99% compared with the baseline. This advancement surpasses recent methods (e.g., RISE and CSTS) and further validates the efficacy of our dual-stage feature diversity optimization strategy.

(2) Unlike traditional rotation-based self-supervised approaches, OFMFS first enforces geometric consistency. The method then imposes frequency-domain consistency, forming a dual-domain collaborative task. The linking of the two strategies increases the difficulty of self-supervised task, enabling the representation to maintain a balance between global outline and local detail.

(3) We observe that the classification accuracy of OFMFS in fine-grained datasets is outstanding, which can be attributed to the cross-domain collaboration in the FMT stage. By jointly modeling geometric features and frequency information, OFMFS preserves the overall structure of the bird and detailed features of the feather edge pattern.

Table 2 reports the results on CIFAR-FS. The following observations can be made from these results.

- (1) As reported in Table 2, OFMFS achieves 73.87% for 5-way 1-shot and 86.97% for 5-way 5-shot on CIFAR-FS. The corresponding gains over the baseline are 0.78% and 1.42%.
- (2) When traditional FSL approaches (e.g., ProtoNet, MAML, and RelationNet) face low-resolution images of the CIFAR-FS, these methods struggle to capture detailed feature representations in this situation. During the FMT stage of OFMFS, we introduce a wavelet subband discrimination task, enabling the model to learn the overall structure and detailed features of the image simultaneously. In addition, the multi-scale characteristic of DWT further enhances the adaptability of OFMFS in dealing with low-resolution images.

The MiniImageNet results are summarized in Table 3, from which we can make the following observations.

- (1) The MiniImageNet comparison results are summarized in Table 3. The classification accuracy of OFMFS increases by 1.36% in the 1-shot setting. In the 5-shot setup, the improvement reaches 3.27%.
- (2) We observe that the classification accuracy of the recent DouN model is close to OFMFS from Table 3. The DouN model is based on a graph neural network, which realizes context-aware feature propagation through node splitting of the graph structure.

Table 1

Accuracy (%) comparison of various FSL and self-supervised methods in 5-way 1-shot and 5-shot experiments on CUB-200-2011.

Method	Backbone	5-way 1-shot	5-way 5-shot
Few-Shot Learning Methods			
MatchingNet [2]	ConvNet	61.16 $\pm$ 0.89	72.86 $\pm$ 0.70
ProtoNet [1]	ConvNet	66.36 $\pm$ 1.00	82.03 $\pm$ 0.59
MAML [29]	ConvNet	66.26 $\pm$ 1.05	78.82 $\pm$ 0.70
RelationNet [30]	ConvNet	64.38 $\pm$ 0.94	80.16 $\pm$ 0.64
CSTS [31]	ConvNet	60.83 $\pm$ 0.45	77.12 $\pm$ 0.44
GPRN [32]	ConvNet	59.21 $\pm$ 0.93	73.42 $\pm$ 0.76
DeepEMD [33]	ResNet-12	79.27 $\pm$ 0.29	89.80 $\pm$ 0.51
DouN [34]	ResNet-12	76.72 $\pm$ 0.41	92.54 $\pm$ 0.23
CADS [35]	ResNet-12	76.47 $\pm$ 0.21	89.45 $\pm$ 0.12
AFHN [36]	ResNet-18	70.53 $\pm$ 1.01	83.95 $\pm$ 0.63
Self-Supervised Learning Methods			
CSS [37]	ConvNet	66.01 $\pm$ 0.90	81.84 $\pm$ 0.59
IEPT [38]	ConvNet	69.97 $\pm$ 0.49	84.33 $\pm$ 0.33
RISE [39]	ConvNet	67.42 $\pm$ 0.96	82.86 $\pm$ 0.59
FS2Net [40]	ConvNet	71.77 $\pm$ 0.85	86.22 $\pm$ 0.53
CPLAE [41]	ResNet-12	69.77 $\pm$ 0.50	84.57 $\pm$ 0.33
Exemplar [42]	WRN-28-10	71.58 $\pm$ 0.32	84.63 $\pm$ 0.57
Baseline++ [43]	WRN-28-10	70.40 $\pm$ 0.81	82.92 $\pm$ 0.78
Rotation [16]	WRN-28-10	75.61 $\pm$ 0.86	87.32 $\pm$ 0.46
S2M2 _R [21]	WRN-28-10	76.12 $\pm$ 0.87	88.62 $\pm$ 0.48
OFMFS (ours)	WRN-28-10	80.49 $\pm$ 0.77	93.61 $\pm$ 0.79

Table 2

Accuracy (%) comparison of various FSL and self-supervised methods in 5-way 1-shot and 5-shot experiments on CIFAR-FS.

Method	Backbone	5-way 1-shot	5-way 5-shot
Few-Shot Learning Methods			
MatchingNet [2]	ConvNet	51.32 $\pm$ 0.85	68.93 $\pm$ 0.74
ProtoNet [1]	ConvNet	55.50 $\pm$ 0.70	72.00 $\pm$ 0.60
MAML [29]	ConvNet	58.90 $\pm$ 1.90	71.50 $\pm$ 1.00
RelationNet [30]	ConvNet	55.00 $\pm$ 1.00	72.00 $\pm$ 0.60
CSTS [31]	ConvNet	62.47 $\pm$ 0.47	81.12 $\pm$ 0.42
GPRN[32]	ConvNet	62.40 $\pm$ 0.92	78.58 $\pm$ 0.70
DeepEMD [33]	ResNet-12	46.47 $\pm$ 0.78	63.22 $\pm$ 0.71
MetaTEDL [44]	ResNet-12	71.30 $\pm$ 0.90	85.20 $\pm$ 0.60
PBML [45]	ResNet-12	73.07 $\pm$ 0.59	85.51 $\pm$ 0.41
AFHN [36]	ResNet-18	68.32 $\pm$ 0.93	81.45 $\pm$ 0.87
MFRL [46]	WRN-28-10	71.70 $\pm$ 0.90	86.20 $\pm$ 0.90
CAML [47]	Vit	71.10 $\pm$ 0.10	85.90 $\pm$ 0.10
Self-Supervised Learning Methods			
CSS [37]	ConvNet	56.49 $\pm$ 0.93	74.59 $\pm$ 0.72
PSST [48]	ConvNet	64.37 $\pm$ 0.33	80.42 $\pm$ 0.32
RISE [39]	ConvNet	60.05 $\pm$ 0.96	78.44 $\pm$ 0.64
SLA [49]	ResNet-12	73.50 $\pm$ 0.70	86.70 $\pm$ 0.50
Boosting [50]	WRN-28-10	73.60 $\pm$ 0.30	86.00 $\pm$ 0.20
Exemplar [42]	WRN-28-10	70.05 $\pm$ 0.17	84.01 $\pm$ 0.22
Baseline++ [43]	WRN-28-10	67.50 $\pm$ 0.64	80.08 $\pm$ 0.32
Rotation [16]	WRN-28-10	70.66 $\pm$ 0.20	84.15 $\pm$ 0.14
S2M2 _R [21]	WRN-28-10	73.09 $\pm$ 0.19	85.55 $\pm$ 0.13
OFMFS (ours)	WRN-28-10	73.87 $\pm$ 0.18	86.97 $\pm$ 0.23

This operation enables the model to better understand the semantic information and contextual relationships of the image. This strategy is similar to the feature blocks random reorganization strategy of the FDL phase. Therefore, they exhibit similar accuracy on MiniImageNet.

Table 3

Accuracy (%) comparison of various FSL and self-supervised methods in 5-way 1-shot and 5-shot experiments on MiniImageNet.

Method	Backbone	5-way 1-shot	5-way 5-shot
Few-Shot Learning Methods			
MatchingNet [2]	ConvNet	43.56 $\pm$ 0.84	55.31 $\pm$ 0.73
ProtoNet [1]	ConvNet	50.37 $\pm$ 0.83	67.33 $\pm$ 0.67
MAML [29]	ConvNet	50.96 $\pm$ 0.92	66.09 $\pm$ 0.71
GPRN [32]	ConvNet	53.42 $\pm$ 0.79	69.42 $\pm$ 0.66
RelationNet [30]	ConvNet	51.84 $\pm$ 0.88	64.55 $\pm$ 0.70
CSTS [31]	ConvNet	62.38 $\pm$ 0.48	79.77 $\pm$ 0.44
DeepEMD [33]	ResNet-12	65.91 $\pm$ 0.82	82.41 $\pm$ 0.56
MetaTEDL [44]	ResNet-12	60.40 $\pm$ 0.80	78.00 $\pm$ 0.60
DouN [34]	ResNet-12	65.87 $\pm$ 0.11	85.13 $\pm$ 0.19
PBML [45]	ResNet-12	63.60 $\pm$ 0.70	81.94 $\pm$ 0.44
AFHN [36]	ResNet-18	62.38 $\pm$ 0.72	78.16 $\pm$ 0.56
MFRL [46]	WRN-28-10	61.83 $\pm$ 0.82	80.26 $\pm$ 0.61
Self-Supervised Learning Methods			
CSS [37]	ConvNet	50.85 $\pm$ 0.84	68.08 $\pm$ 0.73
IEPT [38]	ConvNet	58.43 $\pm$ 0.46	75.07 $\pm$ 0.33
PSST [48]	ConvNet	57.04 $\pm$ 0.51	73.17 $\pm$ 0.48
RISE [39]	ConvNet	53.22 $\pm$ 0.81	69.41 $\pm$ 0.67
FS2Net [40]	ConvNet	64.57 $\pm$ 0.20	81.83 $\pm$ 0.13
SLA [49]	ResNet-12	62.93 $\pm$ 0.63	79.63 $\pm$ 0.47
Boosting [50]	WRN-28-10	63.77 $\pm$ 0.45	80.70 $\pm$ 0.33
Exemplar [42]	WRN-28-10	62.20 $\pm$ 0.45	78.80 $\pm$ 0.15
Baseline++ [43]	WRN-28-10	57.53 $\pm$ 0.10	72.99 $\pm$ 0.43
Rotation [16]	WRN-28-10	63.90 $\pm$ 0.18	81.03 $\pm$ 0.11
S2M2 _R [21]	WRN-28-10	64.88 $\pm$ 0.18	83.18 $\pm$ 0.11
OFMFS (ours)	WRN-28-10	66.24 $\pm$ 0.19	86.45 $\pm$ 0.20

Table 4

Ablation experimental results of OFMFS between different modules on CUB-200-2011 and CIFAR-FS.

Method	CIFAR-FS		CUB-200-2011	
	1-shot	5-shot	1-shot	5-shot
Baseline	73.09 $\pm$ 0.19	85.55 $\pm$ 0.13	76.12 $\pm$ 0.87	88.62 $\pm$ 0.48
+ FMT	73.56 $\pm$ 0.43	85.82 $\pm$ 0.51	77.36 $\pm$ 0.74	89.79 $\pm$ 0.69
+ FDL	72.24 $\pm$ 0.39	84.75 $\pm$ 0.46	75.07 $\pm$ 0.79	88.63 $\pm$ 0.74
OFMFS	73.87 $\pm$ 0.18	86.97 $\pm$ 0.23	80.49 $\pm$ 0.77	93.61 $\pm$ 0.79

4.3. Analysis of different modules

To examine the contributions of FMT and FDL, we conduct ablation studies on CIFAR-FS and CUB-200-2011. The results in Table 4 lead to the following observations.

- (1) On CIFAR-FS, the FMT strategy brings accuracy gains of 0.47% and 0.27% over the baseline in the 5-way 1-shot and 5-way 5-shot settings, respectively. These outcomes indicate that the FMT strategy fully utilizes angle information and frequency-domain information in low-resolution (32×32) scenarios. This operation enables rich discriminative features and provides a better global prior for downstream tasks.
- (2) On CUB-200-2011, the joint use of FMT and FDL yields accuracy gains of 4.37% and 4.99% over the baseline in the 5-way 1-shot and 5-way 5-shot settings, respectively. These results indicate that the synergy between FMT and FDL reflects a multi-granularity feature learning process. FMT establishes global structural anchors, while FDL refines local discriminative boundaries. This combination is particularly beneficial for fine-grained datasets like CUB-200-2011.
- (3) These results indicate that FMT and FDL hardly contribute to the final performance in a simple additive manner. In particular, FDL alone brings limited improvement or even performance degradation because FDL is not designed as an independent feature augmentation module. Without the FMT module, patch-level feature reorganization may disrupt semantic consistency.
- (4) In contrast, FMT first provides a structured representation space by jointly learning rotation and wavelet self-supervised information. Within this representation space, FDL can reorganize local feature patches in a more meaningful way, encouraging the model to capture fine-grained discriminative patterns. Therefore, the performance enhancement of OFMFS mainly comes from the staged dependency between FMT and FDL.

Table 5

Ablation experiment for the FMT stage evaluates the connection between self-supervised tasks on CUB-200-2011 and CIFAR-FS.

Method	CIFAR-FS		CUB-200-2011	
	1-shot	5-shot	1-shot	5-shot
Rot	70.66 $\pm$ 0.20	84.15 $\pm$ 0.14	75.61 $\pm$ 0.86	87.32 $\pm$ 0.46
Wave	70.47 $\pm$ 0.46	83.86 $\pm$ 0.39	74.30 $\pm$ 0.54	86.43 $\pm$ 0.57
Rot + Wave	72.23 $\pm$ 0.37	85.39 $\pm$ 0.24	76.71 $\pm$ 0.48	88.43 $\pm$ 0.72
FMT	73.87 $\pm$ 0.18	86.97 $\pm$ 0.23	77.36 $\pm$ 0.74	89.79 $\pm$ 0.69

4.4. Analysis of the connections between self-supervised tasks

Table 5 compares different self-supervised task combinations on CIFAR-FS and CUB-200-2011 under 1-shot and 5-shot settings. Rot refers to the self-supervised task of predicting rotation angles. Wave refers to predicting frequency bands. Rot + Wave represents parallel training of two self-supervised tasks. FMT is our self-supervised strategy. The strategy involves rotating the image and then applying DWT, jointly predicting the rotation angle and the frequency band. Based on the experimental results, we draw the following conclusions.

- (1) The rotation prediction task performed better than the frequency band prediction task on both datasets when comparing single self-supervised tasks. From the perspective of task characteristics, the Rot task directly learns the global geometric structure of images. In contrast, the Wave task focuses more on local frequency-domain features. From the perspective of task difficulty, the Wave task only predicts four frequency band categories. This operation limits the ability of the model to fully extract semantic features from images.
- (2) Compared with single self-supervised tasks, the Rot + Wave task improves performance on both datasets. From the perspective of task synergy, the Rot task provides global geometric priors, and the Wave task complements local frequency details.
- (3) Our FMT strategy further improves upon Rot + Wave. On CUB-200-2011, accuracy increases by 0.65% in 1-shot and 1.36% in 5-shot classification. On CIFAR-FS, it improves by 1.64% in 1-shot and 1.58% in 5-shot classification. By designing a cascaded dual self-supervised task, the model is guided to learn the joint distribution of geometric and frequency information.

4.5. Analysis of wavelet basis selection

Table 6 compares fixed and adaptive wavelet basis strategies on CUB-200-2011, CIFAR-FS, and MiniImageNet. We analyze their effects under both 1-shot and 5-shot settings.

- (1) Compared with fixed wavelet basis selection, adaptive wavelet basis selection improves classification accuracy in most cases. This strategy allows the model to automatically choose suitable wavelet bases by calculating image edge density and texture complexity. This operation transforms the traditional static feature extraction approach which relies on fixed basis functions into a data-driven dynamic adjustment mechanism.
- (2) We notice that the fixed Haar wavelet basis selection achieves higher accuracy than the adaptive wavelet basis selection method in the 1-shot classification task on CIFAR-FS. CIFAR-FS comprises low-resolution (32×32) images, and discriminative features depend on edge contours instead of intricate textural details. The adaptive wavelet basis selection strategy introduces redundant features, which diminishes the prominence of crucial edge information. As a result, the classification accuracy in the 1-shot setting decreases.

4.6. Analysis of different wavelet subbands selection

To examine how the FMT stage affects downstream classification, we test different wavelet subband configurations. Four rotation angles are fixed and randomly paired with four wavelet subbands. Fig. 4 reports the results on CUB-200-2011 under the 5-way 1-shot and 5-way 5-shot settings.

Table 6

Ablation experiment results of OFMFS analyzing fixed wavelet bases and adaptive wavelet bases on CUB-200-2011, CIFAR-FS, and MiniImageNet.

Wavelet Base	CUB-200-2011		CIFAR-FS		MiniImageNet	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Haar	79.21	91.75	74.05	86.39	64.69	84.46
Daubechies	78.63	90.82	72.41	85.44	65.25	84.76
Symlets	79.84	92.38	73.57	84.76	64.90	85.21
Adaptive wavelet	80.49	93.61	73.87	86.97	66.24	86.45

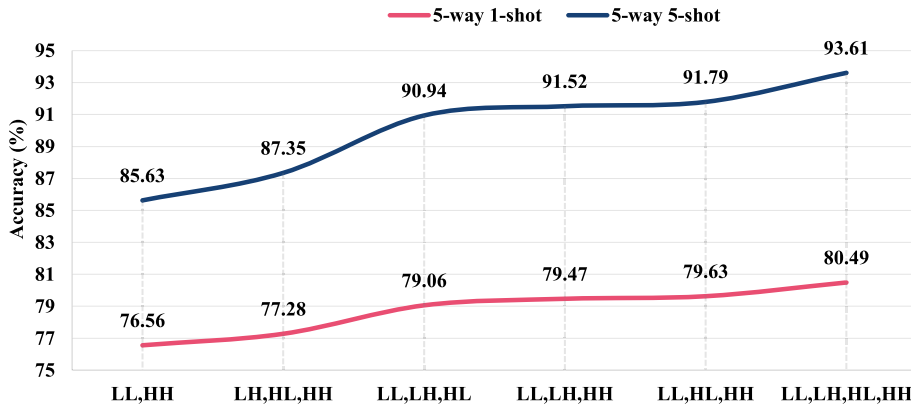


Fig. 4. Classification performance with different wavelet subband selections under 5-way 1-shot and 5-way 5-shot settings on CUB-200-2011.

- (1) The accuracy improves steadily as more suitable wavelet subband combinations are used. Specifically, the performance increases from 76.56% to 80.49% in the 5-way 1-shot setting and from 85.63% to 93.61% in the 5-way 5-shot setting. These gains suggest that wavelet subband selection helps OFMFS capture multi-scale visual information more effectively.
- (2) We observe that the combination including the low-frequency subband (LL, LH, HL) outperforms the high-frequency combination (LH, HL, HH) in both settings. These results indicate that low-frequency information is essential for capturing the main structure of an image in fine-grained scenarios. The low-frequency subband provides prior information about the location and overall structure of the main subject, while the high-frequency subbands emphasize local details. Their combination enables the model to understand the image content more comprehensively.
- (3) Using more wavelet subbands consistently improves the classification results. This suggests that richer frequency information helps OFMFS describe coarse image structures and fine local patterns more effectively. When all wavelet subbands are used, the accuracy reaches 80.49% for 5-way 1-shot and 93.61% for 5-way 5-shot.

4.7. Parameter sensitivity analysis

We examine the effects of ξ and γ on OFMFS. The parameter ξ determines the strength of feature reorganization in FDL, and γ affects the loss weighting in the FMT stage. Experiments are performed on CUB-200-2011 under both 5-way 1-shot and 5-way 5-shot settings.

4.7.1. Sensitivity analysis of ξ

In this section, we discuss the synergy between the parameter ξ (ranging from 0.2 to 0.8) in FMT and FDL. Fig. 5 shows the effectiveness of the ξ parameter on the synergy between FMT and FDL. We perform 5-way 1-shot and 5-way 5-shot experiments on CUB-200-2011, respectively. We have the following observations.

- (1) In Fig. 5, we observe that the highest classification accuracy is achieved when $\xi = 0.5$ in both 1-shot and 5-shot experiments. Compared with extreme parameter settings ($\xi = 0.2$ or $\xi = 0.8$) the improvement is 0.17%–1.07% percentage points. When ξ exceeds 0.7, the model focuses on low-frequency structures, reducing discriminative power for fine-grained details. Conversely, when ξ is below 0.3, excessive local perturbations easily disrupt global semantic consistency. The optimal $\xi = 0.5$ confirms the necessity of integrating multi-granularity features, where global and local features jointly contribute to classification performance.
- (2) As shown in Fig. 5(a), the parameter ξ impacts the performance of OFMFS under the 5-way 1-shot experimental setup. In the case of $\xi = 0.2$ or $\xi = 0.8$, the classification accuracy hardly decreases, indicating that both global and local perspectives can enhance classification when the sample quantity is limited. As shown in Fig. 5(b), the synergistic effectiveness shows stronger

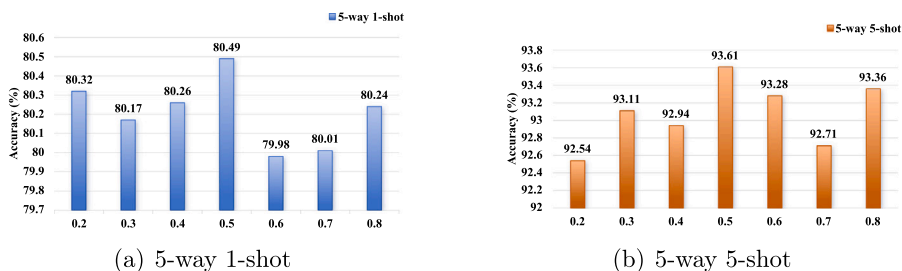


Fig. 5. Influence of the balancing parameter ξ on OFMFS performance on CUB-200-2011.

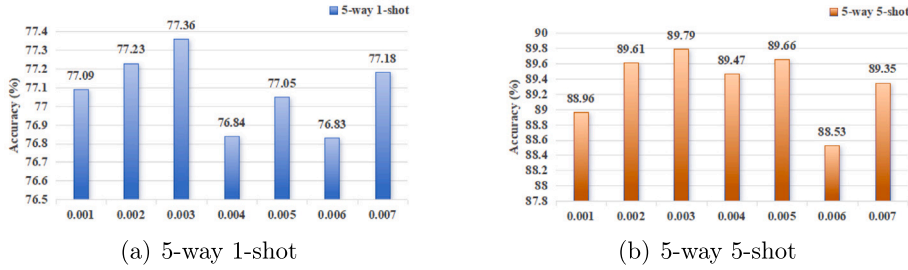


Fig. 6. Effect of γ on 5-way 1-shot and 5-way 5-shot performance on CUB-200-2011.

robustness under the 5-way 5-shot experimental setup. When ξ varies from 0.4 to 0.6, the accuracy ranges are 0.51 and 0.67 percentage points for the 1-shot and 5-shot settings, respectively, indicating relatively stable performance around the optimum. When $\xi = 0.8$, the classification accuracy still reaches 93.36%, demonstrating the importance of the global structural prior in the case of sufficient samples.

- (3) From a cross-stage transfer perspective, the FMT stage extracts global dual-domain information (e.g., object outlines and structures), laying the foundation for subsequent stages. In the FDL stage, the model optimizes the representation of local information through dynamic feature restructuring using the *a priori* knowledge extracted in the FMT stage as a guide. By focusing on fine-grained regions (e.g., beak details, feather edges), the FDL module refines local boundaries that are often decisive for distinguishing visually similar classes. This cross-stage collaboration enhances the understanding of global features and improves classification performance through fine-grained feature refinement.

4.7.2. Sensitivity analysis of γ

We further examine the effect of γ on the balance between supervised classification and self-supervised learning. Fig. 6 reports the corresponding results on CUB-200-2011 under the 5-way 1-shot and 5-way 5-shot settings.

- (1) As shown in Fig. 6, different values of γ lead to different classification results. When $\gamma = 0.003$, OFMFS obtains the highest accuracies of 77.36% and 89.79% in the 5-way 1-shot and 5-way 5-shot settings, respectively. This setting provides a better trade-off between class-label supervision and multi-perspective self-supervised tasks.
- (2) As shown in Fig. 6(a), under the 5-way 1-shot setting, the classification accuracy fluctuates within a relatively narrow range from 76.83% to 77.36%. When γ is too small, such as $\gamma = 0.001$, the adaptive adjustment of the loss weight is relatively weak, which limits the coordination between supervised classification learning and self-supervised representation learning. When γ becomes too large, the balance between the two learning objectives may be disturbed, leading to a slight degradation in classification performance. In contrast, $\gamma = 0.003$ provides a more appropriate compromise, enabling the model to obtain better feature representations.
- (3) As shown in Fig. 6(b), the classification accuracy improves from 88.96% to 89.79% when γ increases from 0.001 to 0.003. A moderate γ enables the model to integrate semantic supervision from the cross-entropy loss and structural information from the self-supervised loss, thereby improving the overall classification performance. Conversely, an unsuitable value of γ weakens this balance and leads to performance degradation.

4.8. Analysis of computational cost and inference time

To quantitatively analyze the computational influence of DWT, we conduct experiments on CUB-200-2011, CIFAR-FS, and MiniImageNet under the 5-way 5-shot setting. In Table 7, Train and Infer denote the training time and inference time per batch, respectively, and both are measured in ms/batch. FLOPs are reported in GFLOPs. We mainly compare two self-supervised methods. One method only predicts the rotation angle. Rot refers to the self-supervised task of predicting rotation angles. The other method is our FMT approach, which predicts both the rotation angle and the wavelet subbands. The DWT operation is only applied during the FMT training stage, and this operation is not performed in the FDL stage or the subsequent testing phases.

Table 7

Computational cost comparison between the self-supervised method that only predicts the rotation angle and FMT (which predicts both the rotation angle and the wavelet subbands) on CUB-200-2011, CIFAR-FS, and MiniImageNet.

Method	CUB-200-2011			CIFAR-FS			MiniImageNet		
	Train	Infer	FLOPs	Train	Infer	FLOPs	Train	Infer	FLOPs
Rotation	198.76	13.60	2312.314	29.40	16.39	335.575	198.82	21.51	2312.314
FMT	898.01	34.98	8671.222	121.11	22.58	1258.455	899.60	38.11	8671.222

- (1) As shown in Table 7, the computational cost of the FMT is affected by the image resolution. On CIFAR-FS, the training time is 121.11 ms/batch and the computational cost is 1258.455 GFLOPs. In contrast, the training time respectively increases to 898.01 ms/batch and 899.60 ms/batch, and the FLOPs reach 8671.222 GFLOPs on CUB-200-2011 and MiniImageNet. This situation confirms that DWT introduces a higher computational cost when handling higher-resolution images, especially during the training stage of FMT.
- (2) To further separate the overhead introduced by DWT from the standard rotation-based self-supervised task, we also compare FMT with the method of predicting rotation angle. Compared with the rotation method, the inference time is 16.39 ms/batch on CIFAR-FS, 13.60 ms/batch on CUB-200-2011, and 21.51 ms/batch on MiniImageNet. After adding DWT, the inference time increases to 22.58, 34.98, and 38.11 ms/batch, respectively.
- (3) The additional inference time overhead caused by DWT remains moderate compared to its training time overhead. This suggests that the additional computational burden introduced by DWT is mainly reflected in the training stage, while the practical inference-time increase over the Rotation-only baseline is relatively limited. Moreover, since DWT is only used in the FMT stage, the final few-shot episodic inference hardly requires additional wavelet decomposition, further reducing the practical inference burden.

4.9. Visualization analysis

We utilize t-SNE and Grad-CAM as two visualization methods to demonstrate the feature representation of OFMFS.

4.9.1. t-SNE visualization analysis of OFMFS

We visualize the features extracted from the baseline, the FMT phase, the FDL phase, and the two-phase collaboration using t-SNE on CUB-200-2011 for the 5-way 5-shot classification task. Fig. 7 presents the visualization of OFMFS, featuring five randomly chosen classes with 50 test samples each, using color coding to distinguish the classes.

Fig. 7(a) shows the feature visualization results of the baseline. We observe that the feature points are dispersed, and there is an obvious overlap between the feature points of different classes. This outcome indicates that the model has limited feature differentiation ability and struggles to differentiate between different classes.

The t-SNE visualization result using only the FMT stage strategy is shown in Fig. 7(b). Compared with the baseline, different classes show separation in the global distribution. This situation is facilitated by the self-supervised task of dual prediction of the rotation angle and wavelet subbands, which enables OFMFS to capture the overall structure of the image better.

Fig. 7(c) shows the FDL stage visualization demonstration. Compared with the baseline, there is an improvement in inter-class separation and a decrease in intra-class aggregation. The feature distribution of FDL exhibits high divergence and local focusing characteristics through dynamic feature reorganization.

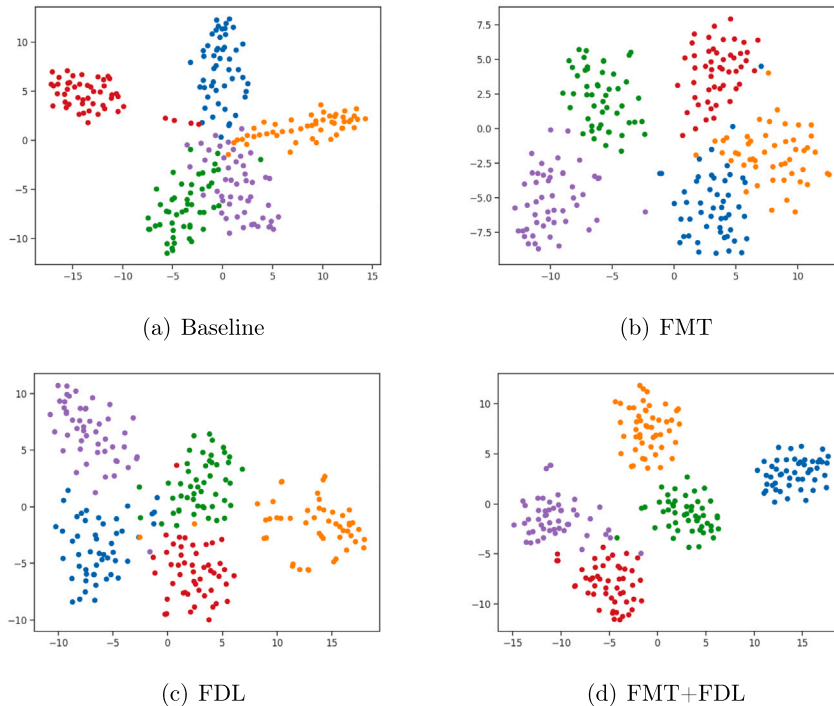


Fig. 7. The t-SNE visualization of Baseline, FMT, FDL, and their collaboration (FMT + FDL) on CUB-200-2011 under the 5-way 5-shot setting. Different colors represent different classes.

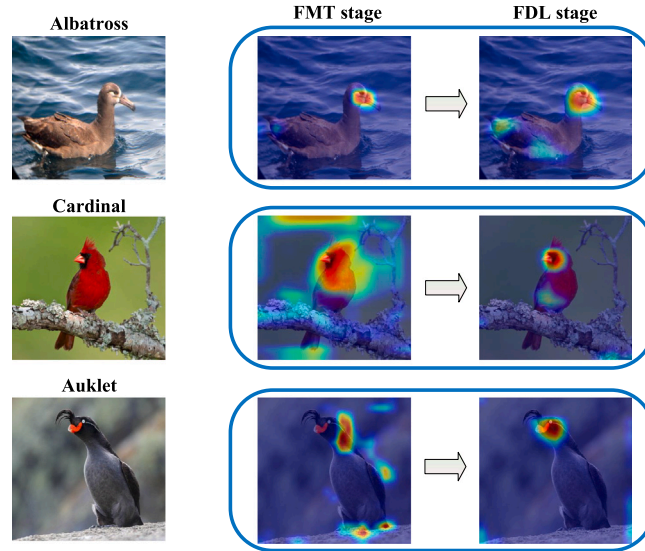


Fig. 8. Grad-CAM visualization analysis of OFMFS for three different fine-grained classes in CUB-200-2011.

The visualization of OFMFS is shown in Fig. 7(d). OFMFS balances global structure and local details through the synergy between FMT and FDL. Compared with FMT-only and FDL-only settings, OFMFS forms more compact intra-class clusters and clearer inter-class boundaries. This suggests that the combination of FMT and FDL improves the structure of the embedding space, supporting the non-additive effect observed in the analysis of different modules.

4.9.2. Grad-CAM visualization analysis of OFMFS

We utilize the Grad-CAM method to visualize the feature regions that the model focuses on during both the FMT and FDL stages. These three images were randomly selected from CUB-200-2011. As shown in Fig. 8, the visualization results demonstrate the key regions of interest identified by the model at different stages, with darker colors indicating higher attention to those regions.

We observe that the Cardinal class has the discriminative features of red feathers and black facial stripes. During the FMT stage, the attention of the model is scattered. In the FDL stage, the attention of the model concentrates on the black facial stripes and the red feathers on the chest. The model learns that red feathers are a common class feature through the FDL stage. The black facial stripes are specific fine-grained features. This change indicates that the focus of the model shifts from global color perception to local discriminative markers.

The heatmap visualization further validates the design logic of the OFMFS. In the FMT stage, multi-perspective self-supervised tasks help build global feature anchors. These anchors guide the model in initially perceiving key visual regions. In the FDL stage, feature decoupling enhances the sensitivity to local details. This operation further directs the model to focus on fine-grained features.

FDL alone tends to produce scattered or unstable activation regions, indicating that local feature reorganization without FMT may introduce noisy attention. In contrast, OFMFS focuses more consistently on discriminative object parts. This supports our explanation that FMT provides dual-domain aware semantic anchors, enabling FDL to enhance local discriminative features rather than generating random perturbations.

5. Conclusion and future work

In this paper, we introduce an optimizing feature diversity via multi-perspective self-supervised tasks (OFMFS). OFMFS combines feature multi-perspective training (FMT) and feature diversity learning (FDL) to address the challenge of limited feature representation in few-shot learning. During the FMT stage, OFMFS employs rotated angle and wavelet subband prediction tasks to learn comprehensive and diverse features. In the FDL stage, OFMFS applies dynamic feature recombination to further enhance the sensitivity of fine-grained features. We conduct a series of experiments on three standard few-shot classification datasets in 1-shot and 5-shot settings to validate the effectiveness of OFMFS. However, the wavelet transform may introduce additional computational costs when processing high-resolution images. In addition, the current work mainly focuses on few-shot image classification, and the generalization of the FMT-FDL synergy to standard image classification tasks still needs further investigation. In the future, we will explore different frequency-domain processing methods for images, and broader validation on standard image classification tasks will be considered.

CRediT authorship contribution statement

Dongqing Li: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Linhua Zou:** Writing – review & editing, Validation, Supervision, Software, Resources, Funding acquisition. **Wencheng Lin:** Writing

– review & editing, Visualization, Validation, Supervision, Software, Investigation, Formal analysis. **Hong Zhao:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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